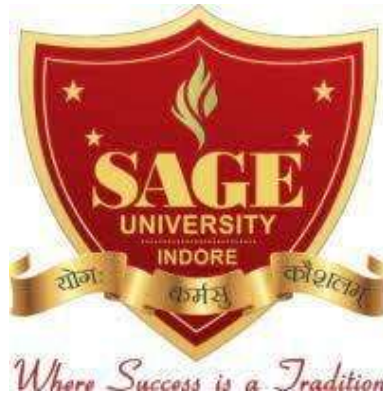


Full Stack Development at Cybershoora Pvt Ltd.

A Internship Report Submitted To



SAGE UNIVERSITY, INDORE

Towards Partial Fulfillment for the Award of

**Bachelor of Technology
in
Computer Science & Technology**

**Submitted By
Aniket Awasthi
(Enroll.No. 24ADV3CST0320)**

**Institute of Advance Computing core
SAGE University
Bypass Road, Kailod Kartal, Indore
Madhya Pradesh - 452020
Aug -2026**

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2. The Aim Of The Internship :

The primary aim of this three-month Full Stack Development internship was to acquire intensive, hands-on experience in engineering enterprise-grade software solutions within a fast-paced, agile industry environment. The objective was to bridge the theoretical concepts of computer science with practical application by designing, developing, deploying, and maintaining scalable web and mobile architectures. Specifically, the internship aimed to build proficiency in cross-platform mobile development, server-side rendering, real-time bidirectional data flow, secure access control, and intuitive UI/UX implementation, thereby solidifying the foundation required for a professional career in full-stack software engineering.

2.1. Abstract :

This comprehensive report documents a three-month Full Stack Development internship completed at Cyber Shoora Pvt Ltd, spanning from December 26th to February 26th. The tenure was characterized by major contributions to three critical, high-impact enterprise systems. The first project, Cyber Shoora Mail, involved engineering a proprietary custom-domain webmail platform featuring an innovative, integrated "Mail Canvas" that allows users to visually format emails using drag-and-drop mechanics. The second, Cyber Shoora Meet, is a highly secure video conferencing application built with strict hardware authorization protocols, session recording capabilities, and a unique administrative feature permitting delayed, remote activation of participant hardware for seamless moderation. The third project, Tipping Clientwork, centered on the digital transformation of a Haridwar-based temple management institute, requiring the development of a modern frontend interface paired with a highly secure, role-based backend administrative dashboard.

Furthermore, the internship extended into mobile application development, utilizing the Flutter framework to port the Cyber Shoora Mail ecosystem to Android, complete with robust background processes for real-time push notifications. The successful architectural design and deployment of these systems demonstrated a high degree of technical proficiency, culminating in the prestigious "Best Employee" award for the internship duration and a public, professional endorsement from the CEO of Cyber Shoora Pvt Ltd.

2.2. Introduction :

In today's rapidly evolving digital landscape, organizations require highly tailored, secure, and efficient communication infrastructures. Cyber Shoora Pvt Ltd operates at the forefront of providing these robust digital solutions. As a Full Stack Development Intern, the role demanded a comprehensive understanding of the entire software development lifecycle (SDLC)—from initial client requirements and UI/UX wireframing to backend database modeling and cross-platform deployment.

The internship provided an immersive experience into modern JavaScript and Dart ecosystems. It required navigating complex state management, optimizing web applications for performance, and ensuring stringent security measures across multiple user roles. By contributing to diverse projects ranging from real-time communication tools to content management systems, this role offered a panoramic view of modern software engineering challenges and the agile methodologies required to overcome them.

3. The Need of The Internship :

Academic coursework provides the essential theoretical framework of data structures, algorithms, and basic application design; however, the transition to a professional software engineer necessitates exposure to production-level codebases and real-world user constraints. This internship was crucial for understanding how complex, scalable systems operate under concurrent user loads. It provided an indispensable environment to practice secure authentication flows, manage non-relational database structures, implement real-time communication protocols, and understand the nuances of deploying cross-platform applications. Furthermore, it cultivated vital soft skills, including requirement analysis, time management, and translating business logic into functional technical specifications.

3.1. Problem Definition :

The internship addressed several complex technical and operational bottlenecks across distinct project domains:

Project 1: ShooraMail (Web Integration):

The Challenge: The company needed a sovereign, proprietary web-based mailing system for internal and client domain routing (e.g., aniket@cybershoora.com).

The Bottleneck: Standard text editors in existing webmail clients are rigid. The primary challenge was engineering an intuitive, highly customizable visual editor—a "Mail Canvas"—that allowed users to format emails with the fluidity of graphic design tools without breaking standard HTML/CSS email rendering protocols across different email clients.

Project 2: ShooraMeet (Secure Video Conferencing):

The Challenge: Existing third-party meeting applications lacked the granular administrative oversight required by the company's specific operational "tribes."

The Bottleneck: The system needed to strictly enforce pre-join hardware (camera/microphone) permissions, preventing unverified access. Furthermore, administrators required a controversial but necessary feature: the ability to remotely activate a participant's camera or microphone to streamline communication, which presented significant privacy, state-synchronization, and user-experience challenges.

Project 3: Tipping Clientwork (Temple Management System):

The Challenge: A temple management institute in Haridwar suffered from an outdated, static digital presence that could not be easily updated by non-technical staff.

The Bottleneck: Delivering a complete UI/UX overhaul while building a secure, role-based dashboard for admins to safely manage website content without compromising database integrity.

Project 4: ShooraMail (Mobile Accessibility):

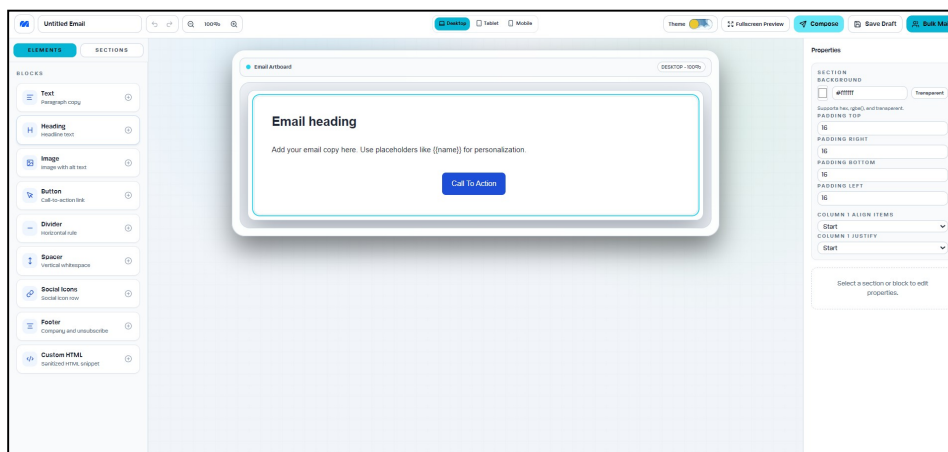
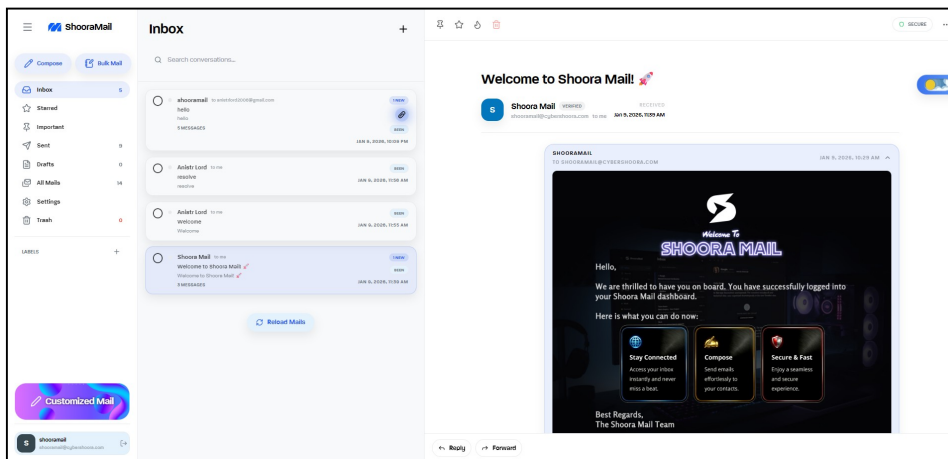
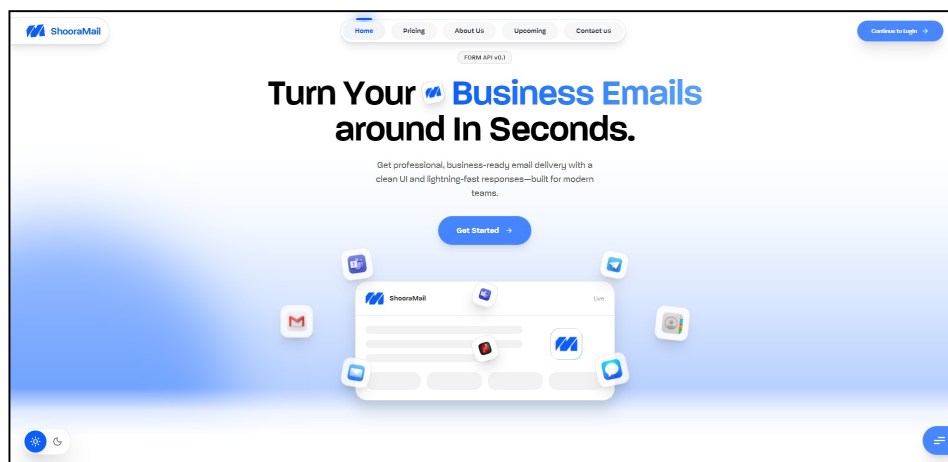
The Challenge: The newly developed web-based mailing system lacked on-the-go accessibility.

The Bottleneck: Users needed to receive instant, reliable push notifications for incoming domain emails even when the application was fully closed or the mobile device was locked, requiring complex background service integration.

3.2. Problem Solution :

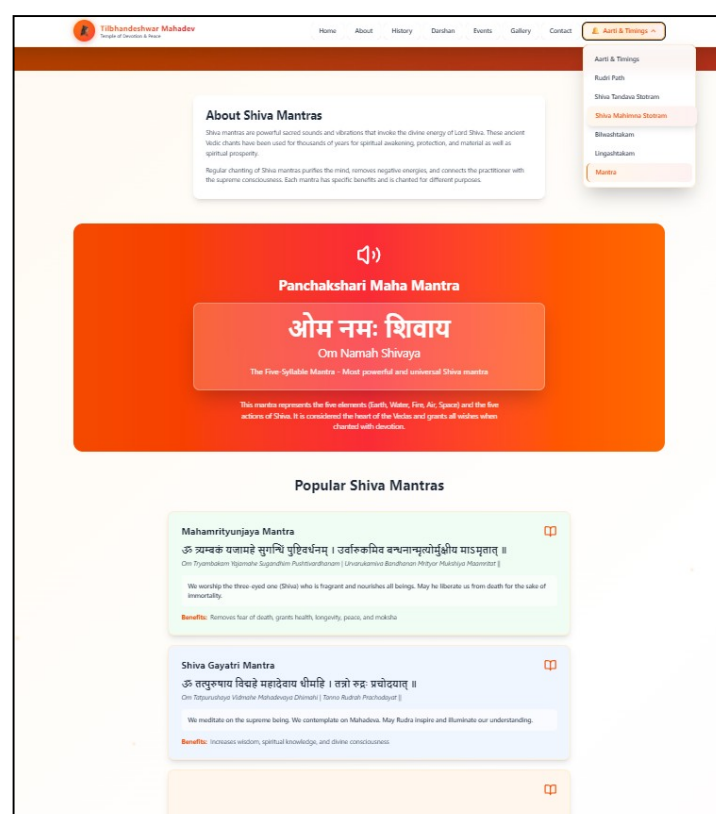
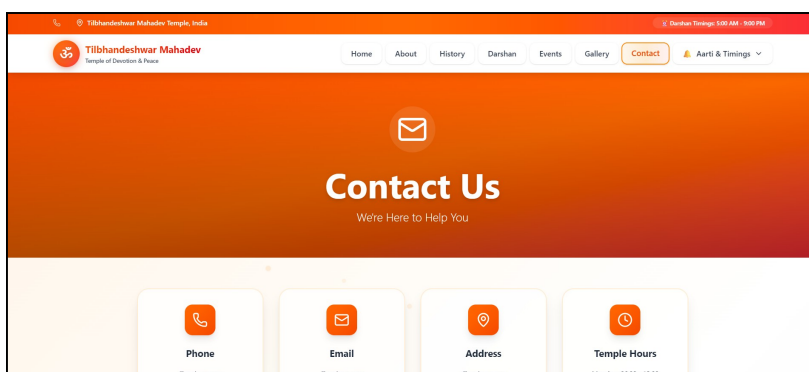
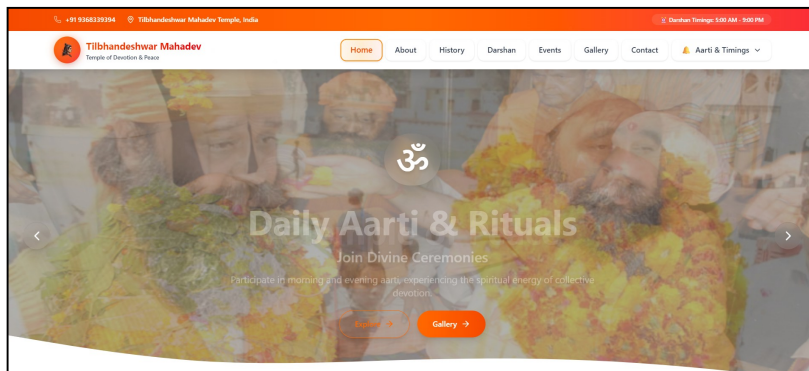
Engineering ShooraMail (Web):

Developed a highly responsive web application utilizing Next.js for server-side rendering, ensuring fast load times and optimized routing. The backend was powered by Node.js to handle mail routing protocols (SMTP/IMAP integrations). To solve the formatting challenge, a custom "Mail Canvas" was engineered. This interactive interface allowed users to drag, drop, and format text and visual elements, which the system then dynamically parsed into clean, universally readable HTML email templates before dispatching.



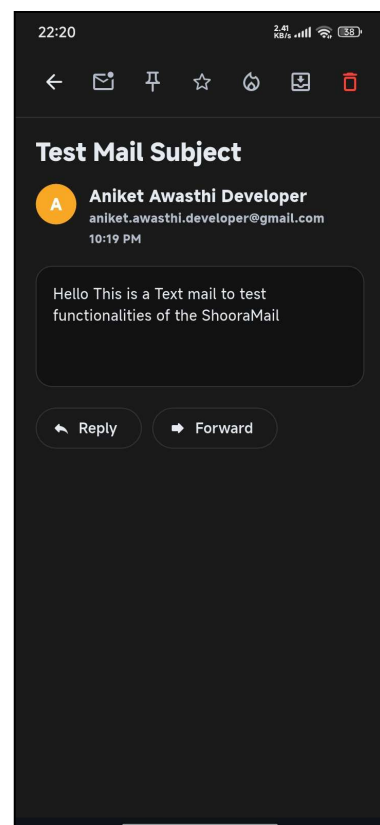
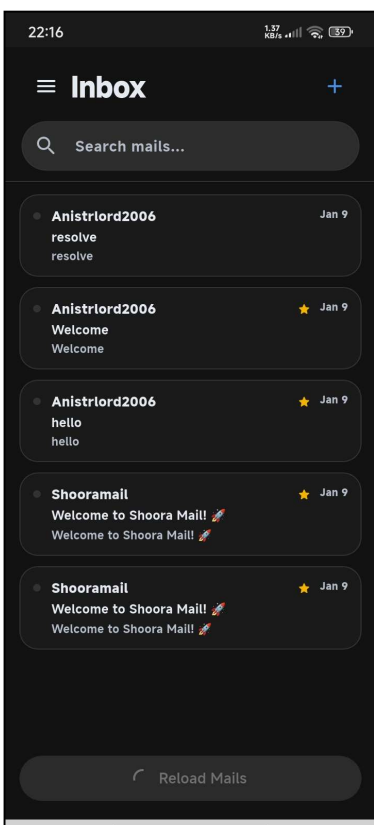
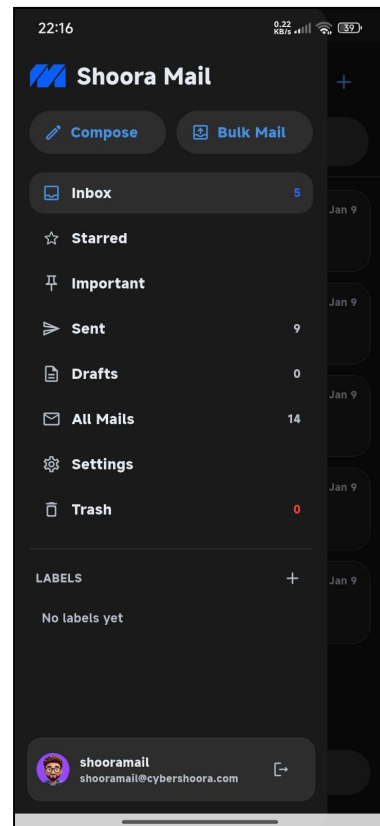
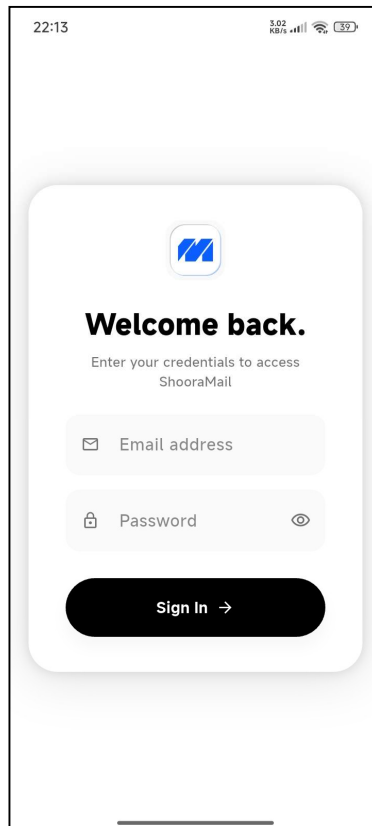
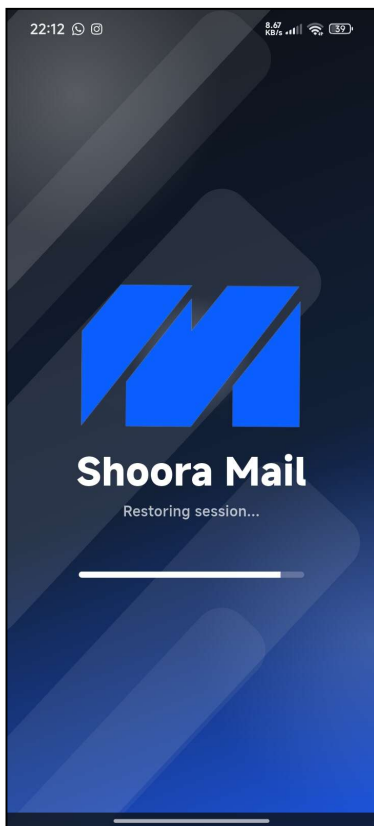
Developing Tipping Clientwork:

Executed a complete digital redesign of the Haridwar temple website. The frontend was built to be highly accessible and visually engaging. For the backend, a robust administration dashboard was constructed. A secure authentication pipeline using hashed passwords and JWT (JSON Web Tokens) was implemented, allowing "Higher Admins" to generate and assign specific roles. This enabled local management to securely update event schedules, temple information, and announcements without directly interacting with the underlying database.



Deploying Shoora Mail (Android App):

Leveraged the Flutter framework to build a natively compiled Android application from a single codebase. The Dart programming language was used to construct a fluid, responsive mobile UI. To solve the notification bottleneck, the app was integrated with cloud messaging services. Background message handlers were implemented to ensure that a persistent listener remained active, instantly pushing device-level notifications for incoming emails even when the app was terminated.



Architecting ShooraMeet:

Built a secure, real-time video conferencing ecosystem. The entry gateway was programmed to perform strict hardware checks, halting the joining process until camera and microphone permissions were explicitly granted at the browser level. For the administrative remote-activation feature, a secure WebSockets connection was established. When an admin triggers a hardware request, a synchronous 4-to-5-second countdown timer initiates on the target user's UI. This buffer ensures user readiness and respects privacy before the application state automatically transitions to un-mute the microphone or enable the camera feed. All sessions were securely routed and recorded with high-level encryption.

4. Technical Descriptions :

4.1. Language Use and Justification :

React & Next.js: Chosen as the primary frontend framework for web applications. Next.js provided crucial Server-Side Rendering (SSR) and Static Site Generation (SSG), which drastically improved the performance, SEO, and initial load times of the applications compared to standard single-page applications.

Node.js: Utilized as the runtime environment for the backend servers. Its non-blocking, event-driven architecture was perfectly suited for handling the high volume of asynchronous requests required by real-time mail routing and video conferencing data streams.

Dart & Flutter: Selected for mobile development. Flutter's widget-based architecture allowed for rapid, highly customizable UI development, while Dart provided the necessary performance and seamless integration with native Android background APIs.

5. System Requirement :

Robust Client-Server Architecture utilizing RESTful APIs for standard data fetching and WebSockets for low-latency, real-time bidirectional communication (vital for the meeting application's timer and hardware control).

Secure Database infrastructure (utilizing NoSQL databases like MongoDB) for flexible, scalable storage of user credentials, hierarchical role data, meeting logs, and dynamic web content.

Integration of Background Notification Services for persistent mobile delivery.

5.1. Hardware Requirement :

Development Environment: High-performance machine with multi-core processors and sufficient RAM to handle local servers, database instances, and mobile Android emulators concurrently.

End-User Environment: Standard modern web browsers (Chrome, Firefox, Edge) with hardware acceleration enabled; Android devices running updated OS versions; functional peripheral hardware (webcams, microphones).

5.2. Software Requirement :

Core Stack: Next.js, React, Node.js, Express, Dart, Flutter SDK

Design & Prototyping: Figma for UI/UX wireframing and layout design prior to coding.

Development Tools: Git and GitHub for version control and collaborative code management; modern, agentic IDEs such as Cursor and Windsurf to accelerate coding workflows, optimize logic, and assist in rapid debugging.

6. Expected Outcomes :

The internship resulted in the successful end-to-end delivery of comprehensive enterprise and consumer-facing applications. The solutions developed directly enhanced Cyber Shoor's internal operational efficiency and provided external clients with modernized, scalable digital infrastructure.

Key Professional Milestones:

Technical Mastery: Developed advanced proficiency in full-stack architecture, successfully bridging frontend UI/UX logic with complex backend database management and real-time state synchronization.

Cross-Platform Competence: Successfully translated complex web functionalities into a seamless, native-feeling mobile experience using Flutter, mastering background task execution.

Industry Recognition: Consistently met aggressive agile development deadlines and exceeded technical requirements, resulting in being officially honored as the "Best Employee" for the internship term.

Professional Endorsement: Received a highly commendable, personalized recommendation from the CEO of Cyber Shoor Pvt Ltd on LinkedIn, publicly validating my capabilities in software engineering, project management, and application deployment.

